PERSONAL INFORMATION:
Name: Dhritiraj Sengupta
Nationality: British
Position: Earth Observation Research Engineer
Current Organisation: GMV NSL, UK

CONTACT DETAILS:
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PROFESSIONAL EXPERIENCE:

2025, May – Present

GMV NSL, UK

Earth Observation Research Engineer

- Implement geospatial foundation models across a range of Earth Observation applications, with particular focus on land use mapping.
- Manage end-to-end EO data processing and delivery, developing structured EO metadata for the ESA APEx, EarthCODE and DeepESDL platforms.
- Lead and contribute to technical proposal writing for ESA and World Bank tenders, coordinating multidisciplinary inputs to secure new business.
- Coordinate bid development across teams and partners, managing proposal timelines, technical inputs and compliance for competitive tenders.

2025-26 Projects –>

1. ESA Earth Observation Best Practices Road and Rail (EOBP)
2. ESA Earth Observation Best Practices Solar Energy (EOBP
3. ESA Earth Observation for Shipping Emission (EO4SEM)
4. ESA Global Development Assistance (GDA) for Public Health, building damage assessment using Sentinel-1 time series analysis
5. Mapping and predicting Macroalgae in the Portuguese coast (EO4RO)
6. Railway vulnerability Mapping using UK Earth Observation Data Hub (EODH)

2023, Sep – April, 2025

Plymouth Marine Laboratory, UK

Earth Observation Scientist in Earth Observation and Science Application group (EOSA) group:

- Manage both technical and non-technical deliverables for the ESA **EO4 Health Resilience project**, overseeing the implementation of Earth Observation (EO) services on **the DeepESDL platform** and leading strategic engagement with key stakeholders.
- Lead co-creation workshops focused on EO and health initiatives, driving collaboration and innovation across multidisciplinary teams.
- **Lead a work package within the ESA WIDGEON project**, utilising citizen science and machine learning algorithms to **enhance land use mapping**. Lead partnership development and management for the ESA-funded EO4 Health Resilience project.
- Work package lead for **Welcome Trust funded project WADIM** to analysis multi-source environmental data to predict Cholera in northern India.
- Coordinate and host technical sessions for the **EO4 Health user forum** 2024 (https://eo4society.esa.int/resources/esa-eo4health-user-forum-2024/), promoting collaboration and knowledge exchange among stakeholders.
- Write policy briefs on Earth Observation applications, supporting the dissemination of key insights and recommendations within the industry.
- Collaborate with PML's International Office to expand global engagement and strategic partnerships.

2022, May – Aug 2023

School of Geography and Environmental Science, University of Southampton, UK

EO Postdoctoral Researcher:

- Post-doctoral Research fellow funded by the Leverhulme trust to map spatial temporal patterns of port and its land use expansion.
- Provide EO hands on training to wide range of audiences via More Than Maps initiatives (https://sartrac.org/resources/more_than_maps/about_us/)
- UK-JAPAN RENKEI FELLOW for research into land use change and methods of mapping between UK and Japan
- British Society of Geomorphology (BSG) grant recipient to equip with Drone training capability for Earth Science research.

2020, Aug – April 2022

East China Normal University, Shanghai, China

EO Postdoctoral Researcher:

- Led a China Science Foundation funded project on mapping artificial land use across various coastal cities globally. Paper published in AGU Earth's Future Journal.
- **Led wetland mapping efforts using drone technology and co-developed Coast Engine,** a cloud-based platform for visualising coastal data across China.

  Contributed to the creation of a global land reclamation map and supported international outreach initiatives as part of the Stake Key Laboratory of Coastal and Estuarine Research (SKELC) team

EDUCATION:

2026 (on-going)	Online certificate course, Applied Agentic AI: Systems, Design & Impact, Virginia Tech
2016 – 2020	PhD in Physical Geography (Natural Sciences), East China Normal University, China. Advanced research and operational experience in mapping artificial coastal land and reclamation dynamics using cloud-native geospatial computing and Earth Observation big data analytics. Skilled in processing large satellite archives (Sentinel, Landsat, SAR datasets) within distributed cloud environments, applying machine learning, time-series analysis, and automated classification workflows for coastal change detection. Led and contributed to global assessments covering 100+ coastal cities, generating high-resolution geospatial products for shoreline evolution, land reclamation quantification, and coastal urban expansion. Results disseminated through high-impact factor peer-reviewed journals, supporting evidence-based coastal management and environmental monitoring.
2018 & 2020	Research Trainee with Prof. Michael Meadows, University of Cape Town, South Africa. Specialising in Delta Landuse change across coastal region of Knysna (South Africa) and Yangtze River (China)
2014 – 2016	Masters in Geography and Geospatial Sciences, Mysore University, India. Project Title: - **Volunteered Geographic Information System (VGI) for tree/vegetation mapping in Mysore City**.
2011 – 2014	BA in Geography, Savitribai Phule Pune University, India

Scientific Engagement and Capacity Building

2021 – Present

- Lead presenter for "Getting started with EO Python with OpenEO" at the UK Earth Observation Conference, 2026.
- Co-led engagement activities to engage professionals from the Railway industry on the mapping of land use change along with other environmental risk to railway track. 2025
- Lead initiatives to promote Future Earth's Coast (FEC) objectives, increasing visibility and growing the coastal research community in Shanghai, China.2022-2025
- Design and execute strategic plans to drive engagement and foster collaboration within the coastal research sector.
- Cultivate partnerships and collaborations to advance FEC's mission and support key organisational goals.
- Built Geospatial capacity for visiting students at University of Cape Town, South Africa as a part of International Geographical Union.
- **Online Webinars:**

  https://www.youtube.com/watch?v=s5QS7C55duA,
  https://www.youtube.com/watch?v=25mg9xKnT1s ,
  https://www.youtube.com/watch?v=BEZVryrW55g ,
  https://www.youtube.com/watch?v=xqyUrPyT0QQ
- **Google's Geo for Good Award : https://pml.ac.uk/news/pmls-dr-dhritiraj-sengupta-recognised-with-google/**

- Recipient of the prestigious UK's Global Talent Visa endorsed by the UK Research and Innovation (2022)

RESEARCH IN THE NEWS

- NASA Science (Earth Observatory), "The Expansion of Shanghai" — feature citing research on rapid coastal land reclamation in Shanghai. https://science.nasa.gov/earth/earth-observatory/the-expansion-of-shanghai-145968/
- AGU Newsroom, "New Land Creation on Waterfronts Increasing, Study Finds" — press release on the Earth's Future paper. https://news.agu.org/press-release/new-land-creation-on-waterfronts-increasing-study-finds/
- The Conversation, "Artificial Developments Weaken Coastal Resilience – Here's How Mapping Them Can Help" – authored feature on coastal reclamation and resilience. https://theconversation.com/artificial-developments-weaken-coastal-resilience-heres-how-mapping-them-can-help-250299
- The Guardian, "Humans have reclaimed 'land size of Luxembourg' since" 2000 https://www.theguardian.com/science/2023/mar/22/humans-have-reclaimed-land-size-of-luxembourg-since-2000
- Earth observation scientist Dr Dhritiraj Sengupta from Plymouth Marine Laboratory explains how data from satellites can help us monitor and predict health and nature issues. https://www.siliconrepublic.com/innovation/satellite-data-health-nature-research-esa

KEY SKILLS

Languages Python (Proficient), MATLAB, R, JavaScript, English (Fluent), Mandarin (HSK-2), Bengali (Mother tongue), Hindi (Fluent), Assamese (Fluent), Spanish (enrolled in course for September 2026)

Computers PC (Windows, Ubuntu Linux), Google Earth Engine, Microsoft Planetary Computers

KEY EXPERTISE

Earth Observation and Geospatial capacity building, Proposal Writing and Bid Leadership, Data science and Machine Learning. UK Drone Pilot and training, Public Speaking, Scientific Research.International Relations

PUBLICATIONS

Over 30 publications in peer-reviewed journals including journals with high impact. Full list available here→ **https://scholar.google.com/citations?user=3ry0vYsAAAAJ&hl=en**

1st Author publications:

1. **Sengupta, D.**, Girbal Brandão, F., Sathyendranath, S., Kulk, G., Conte, A., Ippoliti, C., Candeloro, L., Bucciarelli, M., Moffat, D., Wint, W., Maranesi, M., Scarano, R., Vitorino, J., Brandt, G., & Morbagal Harish, T. (**2025**). Sensing human health from Space: An assessment of applications and big data platforms. Remote Sensing Applications: Society and Environment, 39, 101701. https://doi.org/10.1016/j.rsase.2025.101701
2. **Sengupta, D**. Townsend. D, (**2025**). Pride and prejudice of the Anthropocene coasts. Marine Policy. https://doi.org/10.1016/j.marpol.2025.104463
3. **Sengupta, D.**, & Lazarus, E. D. (**2023**). Rapid seaward expansion of seaport footprints worldwide. Communications Earth & Environment, 4, Article 440. https://doi.org/10.1038/s43247-023-01110-y
4. **Sengupta, D.**, Choi, Y. R., Tian, B., Brown, S., Meadows, M., Hackney, C. R., Banerjee, A., Li, Y., Chen, R., & Zhou, Y. (**2023**). Mapping 21st Century global coastal land reclamation. Earth's Future, 11(2), e2022EF002927. **https://doi.org/10.1029/2022EF002927**
5. **Sengupta, D.**, Chen, R., Meadows, M. E., & Banerjee, A. (**2020**). Gaining or losing ground? Tracking Asia's hunger for "new" coastal land in the era of sea level rise. Science of the Total Environment, 732, 139290. https://doi.org/10.1016/j.scitotenv.2020.139290
6. **Sengupta, D.**, Chen, R., Meadows, M. E., Choi, Y. R., Banerjee, A., & Xia, Z. (**2019**). Mapping trajectories of coastal land reclamation in nine deltaic megacities using Google Earth Engine. Remote Sensing, 11(22), 2621. https://doi.org/10.3390/rs11222621
7. **Sengupta, D.**, Chen, R., & Meadows, M. E. (**2018**). Building beyond land: An overview of coastal land reclamation in 16 global megacities. Applied Geography, 90, 229–238. https://doi.org/10.1016/j.apgeog.2017.12.015